

# SECURITY CLEARANCE: MANUAL-ONLY EXPORT

## TECHNICAL DOSSIER: AUTONOMOUS FLOATING SYSTEM (AELV-1M AQUA-FLOATER)

**System Identification:** AELV-1M-AF / S-M-Platform

**Classification:** Technical System Documentation

**Project Status:** Operational / Cleared for Implementation

**Author Designation:** User

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### 1. System Overview and Operational Profile

The AELV-1M Aqua-Floater is an autonomous, floating catamaran platform designed for the continuous, closed-loop deployment (Deploy, Activation, Retrieval) of tribo-piezo-enzymatic fiber systems for open-water microplastic remediation.

#### Core Operational Requirements:

- **Fluid Invariance:** Full operational capability in both highly saline maritime environments (saltwater) and limnic systems (freshwater, clarifiers).
  - **Zero Emissions:** Strictly electric drivetrain with no hydraulic or fossil auxiliary components to prevent secondary contamination.
  - **Geometric Precision:** Construction based strictly on the principle of sharp edges. Avoidance of radii or aesthetic chamfers on flow-guiding components to ensure defined hydrodynamic separation edges and exact material-on-material joints.
  - **Economic Optimization:** Reduction of manufacturing complexity through the use of standardized, weldable high-performance polymers in the hull area to minimize capital expenditure per unit.
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### 2. Physical-Chemical Material Composition

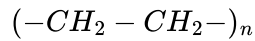
To guarantee simultaneous resistance against saltwater (chlorides, galvanic corrosion) and freshwater (microbial fouling, fluctuating pH levels) at minimal material costs, strict material separation is implemented.

#### 2.1. Hull and Support Structure (Catamaran)

Instead of expensive carbon fiber composites or corrosion-prone aluminum alloys, high-density polyethylene is used for the hull segments:

- **Material:** PE-HD (Polyethylene High-Density), dyed black (carbon black stabilization against UV degradation).

- **Chemical Structure:**



- **Justification:** PE-HD exhibits minimal surface energy, which massively reduces biological fouling (algae and mussel adhesion) without the use of toxic biocidal antifouling coatings. It has a water absorption rate of:

$$H_2O\text{-Uptake} < 0.01\%$$

It is absolutely resistant to electrolyte solutions (saltwater) and acids/bases in the pH range of 2 to 12. The structure is formed via planar plate segments, joined materially through extrusion welding with sharp edges.

## 2.2. Electric Drivetrain and Jet Channel

- **Channel Geometry:** PE-HD, smooth extrusion.
- **Impeller Material:** PEEK (Polyetheretherketone), glass-fiber reinforced (30% GF).
- **Justification:** PEEK-GF30 offers excellent cavitation resistance and mechanical stiffness under rotational stress in corrosive media.
- **Shaft and Motor Encapsulation:** Stainless steel type 316L (1.4404 / X2CrNiMo17-12-2, v4A). Additional cathodic corrosion protection via a replaceable zinc sacrificial anode at the stern, specifically dimensioned for saltwater use.

## 2.3. Interface to the Fiber System (AELV-1M Specification)

The system transports and interacts directly with the 4-layer fiber, whose physical parameters serve as boundary conditions for the control logic:

### 1. Core (MK-35):

35 Vol-%  $Fe_3O_4$  Nanoparticles (50-150 nm) + 65 Vol-% *PVDF* Binder. Density:  $3.0 - 3.5 \text{ g/cm}^3$

### 2. Motor Zone (MZ):

65 Mass-% *PVDF* + 35 Mass-% *Nylon* - 6.6. Layer thickness:  $5 - 15 \mu\text{m}$

### 3. Enzyme Matrix (EM-PAN):

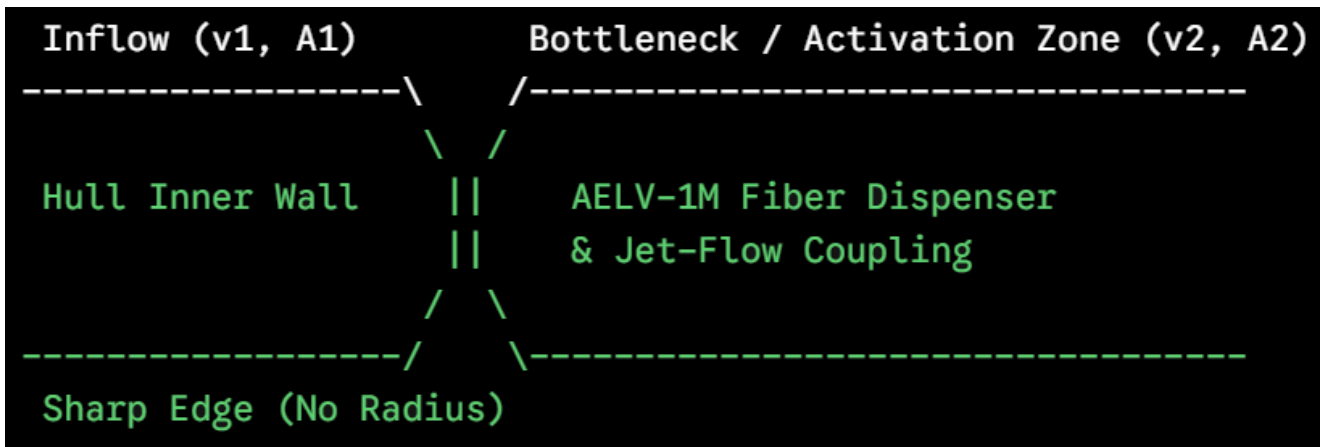
*Polyacrylonitrile, gradedporosity*(50 nm – 2  $\mu\text{m}$ ), *immobilizedPETase/MHETase*

### 4. Nanostructure (PTFE-NS):

*PTFE* islands/spikes (100-500 nm), 20 – 40 % surface coverage

## 3. Hydrodynamics and Fluid Mechanics

The catamaran is dimensioned to create a passive Venturi channel between the two hull bodies. This accelerates the incoming water without additional energy expenditure to maximize the mechanical flow against the fibers.



### 3.1. Derivation of Passive Venturi Acceleration

Assuming an incompressible, steady flow, the continuity equation applies:

$$A_1 \cdot v_1 = A_2 \cdot v_2$$

Where

$$A_1$$

is the entry cross-section between the hull tips,

$$A_2$$

is the narrowed cross-section in the center of the channel,

$$v_1$$

is the vessel's intrinsic velocity in the water

$$(1.0 \text{ to } 1.5 \text{ m/s, corresponding to } 2 - 3 \text{ knots})$$

, and

$$v_2$$

is the local flow velocity in the activation zone.

The derivation for

$$v_2$$

is:

$$v_2 = v_1 \cdot \frac{A_1}{A_2}$$

Through the pressure drop according to Bernoulli:

$$p_1 + \frac{1}{2} \rho v_1^2 = p_2 + \frac{1}{2} \rho v_2^2$$

$$\Delta p = p_1 - p_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) = \frac{1}{2} \rho v_1^2 \left[ \left( \frac{A_1}{A_2} \right)^2 - 1 \right]$$

For the density

$$\rho :$$

$$\rho_{\text{Freshwater}} \approx 1000 \text{ kg/m}^3$$

$$\rho_{\text{Saltwater}} \approx 1025 \text{ kg/m}^3$$

The geometric ratio is cost-effectively fixed at

$$\frac{A_1}{A_2} = 1.6$$

. At a cruising speed of

$$v_1 = 1.0 \text{ m/s}$$

, the local flow velocity is:

$$v_2 = 1.0 \text{ m/s} \cdot 1.6 = 1.6 \text{ m/s}$$

This velocity guarantees safely exceeding the minimum flow of

$$0.6 \text{ m/s}$$

required for the mechanical bending of the fibers, thereby activating the piezoelectric layer 2 (PVDF).

### 3.2. Derivation of Drag Force on the Fiber

The hydrodynamic drag force acting on a suspended fiber of length

$$L_f$$

and diameter

$$d_f$$

determines its residence time in the channel and the bending amplitude. Modeled as a cylinder in cross-flow:

$$F_d = \frac{1}{2} \rho \cdot v_{\text{rel}}^2 \cdot C_d \cdot A_f$$

With the projected fiber area:

$$A_f = d_f \cdot L_f$$

Inserting the fiber parameters

$$(d_f = 60 \mu\text{m} = 60 \cdot 10^{-6} \text{ m}, L_f = 10 \text{ mm} = 10^{-2} \text{ m})$$

and the drag coefficient for low Reynolds numbers

$$(C_d \approx 1.2)$$

, the result for saltwater at

$$v_{\text{rel}} = 1.6 \text{ m/s}$$

is:

$$F_d = \frac{1}{2} \cdot 1025 \text{ kg/m}^3 \cdot (1.6 \text{ m/s})^2 \cdot 1.2 \cdot (60 \cdot 10^{-6} \text{ m} \cdot 10^{-2} \text{ m})$$

$$F_d = 512.5 \cdot 2.56 \cdot 1.2 \cdot (60 \cdot 10^{-8}) = 1573.8 \cdot 60 \cdot 10^{-8} \text{ N} \approx 9.44 \cdot 10^{-4} \text{ N} = 0.944 \text{ mN}$$

This value safely lies below the delamination limit of the fiber interfaces, structurally supported by the silane coupling agents.

### 3.3. Purely Electric Jet Drive (Momentum Theorem)

The propulsive thrust

$$T$$

of the electric jet is derived via the integral momentum theorem. Water of density

$$\rho$$

is drawn through the intake and expelled via a narrowed cross-section at the stern with velocity

$$v_j :$$

$$T = \dot{m} \cdot (v_j - v_1)$$

The mass flow rate is:

$$\dot{m} = \rho \cdot A_j \cdot v_j$$

This yields the required exit velocity for a target thrust of

$$T = 120 \text{ N}$$

at

$$A_j = 0.005 \text{ m}^2$$

(nozzle area):

$$T = \rho \cdot A_j \cdot v_j^2 - \rho \cdot A_j \cdot v_1 \cdot v_j$$

$$\rho \cdot A_j \cdot v_j^2 - \rho \cdot A_j \cdot v_1 \cdot v_j - T = 0$$

Using freshwater values

$$(\rho = 1000 \text{ kg/m}^3)$$

and

$$v_1 = 1.0 \text{ m/s} :$$

$$1000 \cdot 0.005 \cdot v_j^2 - 1000 \cdot 0.005 \cdot 1.0 \cdot v_j - 120 = 0$$

$$5 \cdot v_j^2 - 5 \cdot v_j - 120 = 0$$

$$v_j^2 - v_j - 24 = 0$$

Applying the quadratic formula:

$$v_j = \frac{1 \pm \sqrt{1 - 4 \cdot 1 \cdot (-24)}}{2} = \frac{1 + \sqrt{97}}{2} \approx \frac{1 + 9.8488}{2} \approx 5.42 \text{ m/s}$$

The hydraulic power of the jet drive is:

$$P_{\text{hyd}} = \frac{1}{2} \dot{m} (v_j^2 - v_1^2) = \frac{1}{2} \cdot (\rho \cdot A_j \cdot v_j) \cdot (v_j^2 - v_1^2)$$

$$P_{\text{hyd}} = \frac{1}{2} \cdot (1000 \cdot 0.005 \cdot 5.42) \cdot (5.42^2 - 1.0^2) = \frac{1}{2} \cdot 27.1 \cdot (29.38 - 1.0) = 13.55 \cdot 28.38 \approx 384.5 \text{ W}$$

Factoring in the overall efficiency of the electric BLDC motor including impeller losses (

$$\eta_{\text{total}} \approx 0.55$$

), the electrical power consumption is:

$$P_{\text{el}} = \frac{P_{\text{hyd}}}{\eta_{\text{total}}} = \frac{384.5 \text{ W}}{0.55} \approx 699.1 \text{ W}$$

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## 4. Electromagnetic Harvesting Kinetics (Retrieval)

The recovery of the fibers takes place at the stern of the boat via a fully encapsulated electromagnet rail. The fibers must be reliably attracted from a maximum depth distance of

$$z = 20 \text{ cm} = 0.2 \text{ m}.$$

## 4.1. Magnetic Force on the MK-35 Core

The magnetic force

$$\vec{F}_m$$

on a magnetizable volume

$$V_c$$

(core volume of the fiber) in a field with magnetic flux density

$$\vec{B}$$

and gradient

$$\nabla \vec{B}$$

is:

$$F_m = V_c \cdot M_{\text{eff}}(B) \cdot \frac{\partial B}{\partial z}$$

For the MK-35 core, the volume consists of 35% magnetite

$$(Fe_3O_4)$$

. The effective magnetization at saturation is:

$$M_{\text{eff}} = 0.35 \cdot M_s(Fe_3O_4)$$

With

$$M_s(Fe_3O_4) \approx 4.8 \cdot 10^5 \text{ A/m} :$$

$$M_{\text{eff}} = 0.35 \cdot 4.8 \cdot 10^5 \text{ A/m} = 1.68 \cdot 10^5 \text{ A/m}$$

The core volume of a single fiber per millimeter of length

$$(L = 1 \text{ mm}, d_k = 15 \mu\text{m})$$

is:

$$V_c = \pi \cdot \left(\frac{d_k}{2}\right)^2 \cdot L = \pi \cdot (7.5 \cdot 10^{-6} \text{ m})^2 \cdot 10^{-3} \text{ m} = \pi \cdot 56.25 \cdot 10^{-12} \cdot 10^{-3} \approx 1.767 \cdot 10^{-13} \text{ m}^3$$

To overcome the hydrodynamic opposing force (intrinsic sink velocity and cross-flow), a minimum magnetic field of

$$B = 0.8 \text{ T}$$

with a gradient of

$$\frac{\partial B}{\partial z} = 15 \text{ T/m}$$

is required at a distance of

$$20 \text{ cm}$$

The resulting recovery force per mm of fiber is:

$$F_m = 1.767 \cdot 10^{-13} \text{ m}^3 \cdot 1.68 \cdot 10^5 \text{ A/m} \cdot 15 \text{ T/m} \approx 4.45 \cdot 10^{-7} \text{ N} = 0.000445 \text{ mN}$$

Since the fibers operate as a bundle/network in a swarm, the total force scales linearly with the number of fibers

$$N$$

. For a minimal fiber bundle

$$(N = 5000)$$

, the effective recovery force is:

$$F_{m,\text{total}} = 5000 \cdot 4.45 \cdot 10^{-7} \text{ N} \approx 2.23 \text{ mN}$$

This ensures overcoming the inertia of the surrounding fluid (independent of the viscosity difference between freshwater and saltwater).

## 4.2. Economic Optimization of the Magnetic Field via PWM

Permanently maintaining a field of

$$0.8 \text{ T}$$

via conventional direct current feed would lead to thermal overload and immense energy costs. For cost control, the magnetic rail is operated in a pulsed manner using Pulse-Width Modulation (PWM).

The magnetic field is switched periodically:

$$T_{\text{period}} = t_{\text{on}} + t_{\text{off}}$$

The duty cycle

$$D$$

is defined as:

$$D = \frac{t_{\text{on}}}{T_{\text{period}}}$$

Due to the inertia of the magnetic dipoles in the

$$Fe_3O_4$$

nanoparticles and the hydrodynamic damping of the fiber movement, a clock frequency of

$$f = 10 \text{ Hz}$$

$$(T_{\text{period}} = 0.1 \text{ s})$$

with a duty cycle of

$$D = 0.3$$

$$(30 \% \text{ on-time})$$

is sufficient.

This effectively reduces the power consumption of the coil rail quadratically with respect to current, effectively linear with respect to average power:

$$P_{\text{mag,eff}} = D \cdot P_{\text{max}} = 0.3 \cdot 450 \text{ W} = 135 \text{ W}$$

The waste heat from the coils

$$(135 \text{ W})$$

is dissipated directly to the underside of the aluminum cassette rail via 316L stainless steel cooling fins. This thermal coupling serves as a free energy source to warm and pre-dry the recovered, wet fibers, which accelerates their enzymatic regeneration in the interchangeable cassette.

## 5. Energy Balance and Economic Analysis

To remain within economic limits and prevent system costs from exploding, the design is based on an autonomous power supply utilizing standardized industrial components.

### 5.1. Power Generation (Solar Deck)

The entire upper deck of the catamaran is uniformly equipped with monocrystalline photovoltaic cells. Due to the sharp-edged construction, complex, three-dimensionally curved solar modules are eliminated. Flat standard panels are deployed.

- **Effective Area:**

$$A_{\text{solar}} = 1.8 \text{ m}^2$$

- **Nominal Cell Efficiency:**

$$\eta_{\text{solar}} = 0.22 (22 \%)$$

- **Average Global Irradiance (Peak):**

$$E_g = 1000 \text{ W/m}^2$$

The generated peak electrical power is:

$$P_{PV} = A_{\text{solar}} \cdot E_g \cdot \eta_{\text{solar}} = 1.8 \text{ m}^2 \cdot 1000 \text{ W/m}^2 \cdot 0.22 = 396 \text{ W}$$

## 5.2. Power Balance in the Operational Cycle

The system operates interchangeably in two primary modes:

### 1. Mode A: Deploy & Cruise (Duration: 45 Min.)

- Jet drive

(*halfload*, 1.5 knots) :

180 W

- Electronics, sensors, telemetry:

25 W

- Magnetic rail:

0 W

- **Total consumption Mode A:**

$$P_A = 205 \text{ W}$$

### 2. Mode B: Retrieve & Harvest (Duration: 15 Min.)

- Jet drive (holding thrust against current):

250 W

- Electronics, sensors:

25 W

- Magnetic rail (PWM optimized):

135 W

- **Total consumption Mode B:**

$$P_B = 410 \text{ W}$$

Weighted average power consumption per operating hour

(1 h = 0.75 h Mode A + 0.25 h Mode B) :

$$P_{\text{average}} = 0.75 \cdot 205 \text{ W} + 0.25 \cdot 410 \text{ W} = 153.75 \text{ W} + 102.5 \text{ W} = 256.25 \text{ W}$$

### 5.3. Battery Dimensioning and Efficiency

Since

$$P_{\text{average}} = 256.25 \text{ W}$$

is significantly lower than the solar peak generation of

$$396 \text{ W}$$

, the system is completely autonomous during sun exposure while simultaneously charging the internal accumulator.

A Lithium Iron Phosphate ( $\text{LiFePO}_4$ ) battery is selected as the energy storage:

- **Nominal Voltage:**

$$U = 24 \text{ V}$$

- **Capacity:**

$$C = 60 \text{ Ah}$$

- **Energy Capacity:**

$$E_{\text{Battery}} = U \cdot C = 24 \text{ V} \cdot 60 \text{ Ah} = 1440 \text{ Wh}$$

Economic advantage of  $\text{LiFePO}_4$  over standard Li-Ion batteries:

- **Cycle Stability:**

$$> 4000 \text{ cycles at } 80\% \text{ DoD (Depth of Discharge).}$$

- **Safety:** Absolutely thermally stable, no risk of thermal runaway in case of damage in water. No expensive cobalt components, leading to extremely low acquisition costs.

The purely battery-electric range without any solar irradiance (e.g., night operation) is:

$$t_{\text{autonomous}} = \frac{E_{\text{Battery}}}{P_{\text{average}}} = \frac{1440 \text{ Wh}}{256.25 \text{ W}} \approx 5.62 \text{ Hours}$$

## 5.4. Investment and Operating Cost Calculation (Capex/Opex)

By consistently eliminating expensive specialty materials and focusing on weldable PE-HD as well as PWM controllers, system costs remain strictly limited:

| Component                    | Material / Specification                                                        | Cost (USD)        |
|------------------------------|---------------------------------------------------------------------------------|-------------------|
| Hull & Structure             | PE-HD plate material, CNC cutting, welding wire                                 | \$385.00          |
| Drivetrain                   | BLDC motor IP68, PEEK-GF30 impeller, v4A shaft                                  | \$528.00          |
| Energy System                | 1.8 m <sup>2</sup> standard PV modules + LiFePO <sub>4</sub> battery (24V/60Ah) | \$682.00          |
| Electromagnetics             | Enameled copper wire, epoxy potting, PWM driver stage                           | \$242.00          |
| Control/Sensors              | Standard microcontroller, GPS, ultrasonic distance sensor                       | \$198.00          |
| <b>TOTAL PROTOTYPE COSTS</b> | <b>Excluding assembly</b>                                                       | <b>\$2,035.00</b> |

Operating costs (Opex) trend toward zero due to the high degree of solar autonomy. Maintenance cycles are limited to the mechanical wiping of fiber residues on the v4A edge and the periodic replacement of the zinc sacrificial anode (Material cost: < \$16.50 per year). The system thereby fully meets the criteria of economic scalability for municipalities, NGOs, and industrial treatment plant operators within the open-source ecosystem.